

6.3 What opportunities are there through using smart materials, e-textiles and technical textiles within products?

- a. Demonstrate an understanding of how smart materials change the functionality of products, such as:
- colour changes using thermochromic, photochromic and electrochromic fibres
 - shape-shifting such as shape memory alloy
 - breathable membranes like Gore-Tex.



- b. Understand and recognise how e-textiles are innovative, wearable textiles that incorporate conductive fibres or elements directly into the textile itself, integrating functional performance into products. Consider developments, such as:
- a range of conductive threads and pigments
 - fibretronics
 - a range of programmable controllers using a range of sensors.



- c. Understand how technical textiles are developed for a range of industry sectors, such as:
- i. geotextiles used in civil engineering, coastal engineering and the construction industry
 - ii. the development of fabrics for hi-tech clothing.



7. Manufacturing processes and techniques

Considerations

*Maths &
Science*